

Identifying Obstacles to Equitable Flood Mitigation Funding

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Abstract

As damages from floods, fires, and other climate extremes mount, public investments in risk reduction have also grown. In the US, public spending on risk reduction efforts totaled over \$X in 2024, and demand for such funds regularly exceeds available funding. Evidence suggests that in this competitive landscape for funding, urban and whiter communities have fared better at obtaining funding but are often spending it on more disadvantaged neighborhoods within them. However, the reasons for this trend are unclear: complexities in the application process, cost-benefit analysis requirements, and flood damage may all play a role in driving the uneven funding patterns. Here, using a novel dataset tracking properties from flood exposure to application to funding receipt, we identify how each hurdle in the mitigation process shapes the properties that ultimately benefit. Our findings suggest that application for federal flood mitigation assistance serves as a pinch point where homeowners in less-white neighborhoods and more affordable homes relative to their neighbors apply. This trend deepens for those who ultimately receive funding, while higher flood-exposure has minimal correlation, implying that funding may not reach all who need it or may be interested.

Keywords: Flood mitigation, Climate resilience

1. Introduction

Impacts from natural hazards in the United States are increasing in cost and frequency ([NOAA National Centers for Environmental Information, 2024](#)).

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While demand and funding for climate resilience measures has grown significantly in recent years, there are widespread concerns about access to those funds and the extent to which they perpetuate or mitigate existing inequities (FEMA, 2023; Junod, 2021; Miller, 2023). Evidence suggests that while vulnerable groups including mobile home residents and racial minorities are over-represented in flood hotspots, wealthier and whiter communities have received more federal funds to adapt to and recover from floods (Elliott et al., 2020; Tate et al., 2021). Several explanations for this disparity have been proposed, including federal cost-benefit analysis requirements, limited local government capacity to apply for grants, and lack of household awareness about funding opportunities. However, research to pinpoint funding obstacles has been constrained by only observing who ultimately receives funding, not who is eligible or who applies for funding. As a result, it is unclear what the barriers are, and therefore which policy or procedural changes would most improve access to climate resilience funds.

Here, we combine novel datasets on over two decades of flood exposure and applications for federal funding in North Carolina to examine the full process: who floods, who applies for mitigation funding, and who receives it. This flooded-to-funded pipeline (Fig. 1) is critical for understanding the process by which households experience the Hazard Mitigation Assistance (HMA) process. By analyzing the household- and neighborhood-level characteristics at each stage of the process, we provide new evidence regarding how the application process shapes the distribution of funding. For example, if households located in marginalized communities rarely submit applications, the primary issue may be lack of awareness or resources for applying. In contrast, if they are applying but not being funded, the selection criteria may be driving disproportionate access.

Our initial results demonstrate that the application stage is crucial for understanding the mitigation funding pipeline. 67.5% of properties that experience flooding are eligible for public assistance, and 61% are in communities where local governments have submitted applications for assistance. However, only 2.3% of all flooded properties apply and ~1% are funded. Properties that apply are significantly more likely to have experienced more than one flooding event over the period. Compared to the population of flooded properties, properties that apply for funding have lower property values (58th percentile to 42nd percentile) and come from census block groups with a lower share of white residents (74% to 64%). Among properties receiving funding, property values and the share of white residents in the block group are even lower. These results are consistent with some nationwide findings, where wealthier and whiter communities have been linked to mitigation access (Mach et al., 2019; Elliott et al., 2020), however we add fidelity in that applying and funded properties are also lower-in-value than their flooded neighbors. Notably, these trends emerge at the newly-visible application stage and widen for those ultimately funded. We are the first to demonstrate that unpacking and examining this stage is pivotal to understanding potential obstacles to equitable mitigation funding.

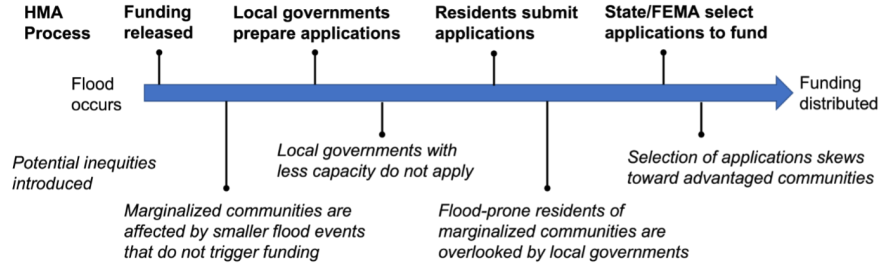


Figure 1: Inequitable outcomes across the FEMA Hazard Mitigation Assistance (HMA) application pipeline.

1.1. Access to federal funding for climate resilience

The federal government plays a central role in supporting disaster recovery and risk mitigation in the United States, so ensuring that federal assistance is reaching those who need it most is crucial. Across the federal level, many entities are involved at different stages in the disaster risk management cycle, including the US Army Corps of Engineers, the National Oceanic and Atmospheric Administration, and the Environmental Protection Agency. However, we focus our discussion and analysis on the Federal Emergency Management Agency, which is the lead entity responsible for disaster response and also a primary funder of risk reduction projects across the US.

There have been long-standing concerns about access to FEMA assistance in the immediate aftermath of disasters. Once a presidential disaster declaration is issued to authorize individual assistance, FEMA operates several programs to provide assistance with immediate post-disaster needs. While these programs are no replacement for insurance, they do provide up to about \$44k for funds for temporary housing, repair or replacement of personal property, unemployment assistance, and transportation or accessibility needs. Racial and income disparities in accessing this post-disaster government assistance have been documented for decades, including the 1996 Northridge earthquake (Bolin and Stanford, 1998), Hurricane Katrina (xx), Hurricane Dolly in 2008 (Rivera, 2022), and the Thomas Fire in 2017 (Méndez et al., 2020). Many reasons for these disparities have been documented, including informational barriers, language challenges, biased home inspections, and other hurdles in the application process.

More recently, concerns have emerged about access to FEMA’s risk mitigation programs. These fund efforts to reduce long-term risk, rather than recover in the immediate aftermath of an event. Typical projects include home elevations, construction of tornado shelters, and property buyouts. More recently, under the Building Resilient Infrastructure and Communities Program (BRIC), FEMA has also funded larger, community-wide risk mitigation projects.

Property-scale projects (e.g., home elevations, property buyouts) are typically

funded under the Hazard Mitigation Assistance Program (HMA), which includes several programs within it: Flood Mitigation Assistance (FMA), Pre-Disaster Mitigation (PDM), and the Hazard Mitigation Grant Program (HMGP). These programs have their own specific eligibility and selection criteria, but all provide funds for property-level flood mitigation. [insert sentence on budget of HMA programs.]

A number of studies have raised concerns about the equity of home elevations and property buyouts, in which a homeowner sells their home to the government for it to be returned to open space. Nationwide analyses find that more urban and whiter counties were more likely to receive funding for buyouts, perhaps because of their capacity to apply for and administer grants (Mach et al., 2019; Elliott et al., 2020). Within those counties, however, communities of lower socioeconomic status and a higher share of racial minorities were more likely to be the locations of buyouts. Home elevation funding has often gone to communities with median incomes over \$100k or are over 90% white (Frank, 2022).

1.2. Procedural challenges to accessing funding

While these studies have presented a number of reasons for funding disparities, pinpointing the chief obstacles has been elusive because of a lack of information on applications for funding, making it impossible to see who is not being selected vs. not being considered in the pool at all. The process for obtaining HMA funding is complicated. As shown in Figure 1, FEMA first releases the funding opportunity. State and local governments then work on developing potential projects by contacting eligible residents, collecting documentation on the property, and conducting a cost-benefit analysis of the project. Local governments (the “sub-applicants”) submit project information to the state (the “applicant”). States may or may not select a subset of projects to then submit to FEMA. FEMA reviews projects, then notifies states and local governments which projects are funded. The federal government typically covers 75% of project costs, with the remaining 25% either falling to some combination of the resident, local government, and state government.

Disparities in access to the funds can arise at many points in this process. Some states provide more support in the project development process than others. Some local governments may choose not to participate at all, due to capacity or financial constraints – thus making it impossible for their flood-prone residents to access the funds. Notably, North Carolina covers the 25% non-federal match required, making it more accessible for cash-strapped local governments. Others local governments may be willing, but do little to no outreach to residents, such that only highly aware and connected residents benefit. The cost-benefit analysis, a mandatory part of project evaluation, may also skew selection.

Our analysis leverages new address-level data on the project development and application process to identify how each stage in this process shapes the distribution of funding. First, we...

2. Bibliography styles

With this template using `elsevier` class, `natbib` will be used. Three bibliographic style files (*.bst) are provided and their use controlled by `cite-style` option:

- `citestyle: number` (default) will use `elsarticle-num.bst` - can be used for the numbered scheme
- `citestyle: numbername` will use `elsarticle-num-names.bst` - can be used for numbered with new options of `natbib.sty`
- `citestyle: authoryear` will use `elsarticle-harv.bst` — can be used for author year scheme

This `citestyle` will insert the right .bst and set the correct `classoption` for `elsarticle` document class.

Using `natbiboptions` variable in YAML header, you can set more options for `natbib` itself. Example

```
natbiboptions: longnamesfirst,angle,semicolon
```

2.1. Using CSL

If `cite-method` is set to `citeproc` in `elsevier_article()`, then `pandoc` is used for citations instead of `natbib`. In this case, the `cs1` option is used to format the references. By default, this template will provide an appropriate style, but alternative `cs1` files are available from <https://www.zotero.org/styles?q=elsevier>. These can be downloaded and stored locally, or the url can be used as in the example header.

3. Equations

Here is an equation:

$$f_X(x) = \left(\frac{\alpha}{\beta}\right) \left(\frac{x}{\beta}\right)^{\alpha-1} e^{-\left(\frac{x}{\beta}\right)^\alpha}; \alpha, \beta, x > 0.$$

Inline equations work as well: $\sum_{i=2}^{\infty} \{\alpha_i^\beta\}$

4. Figures and tables

Figure 2 is generated using an R chunk.

```
plot(runif(25), runif(25))
```

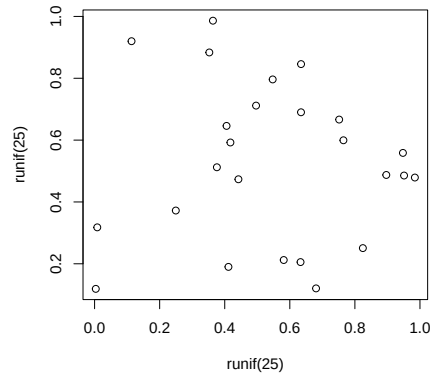


Figure 2: A meaningless scatterplot

5. Tables coming from R

Tables can also be generated using R chunks, as shown in Table 1 example.

```
knitr::kable(head(mtcars)[,1:4])
```

Table 1: Caption centered above table

	mpg	cyl	disp	hp
Mazda RX4	21.0	6	160	110
Mazda RX4 Wag	21.0	6	160	110
Datsun 710	22.8	4	108	93
Hornet 4 Drive	21.4	6	258	110
Hornet Sportabout	18.7	8	360	175
Valiant	18.1	6	225	105

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